• VIBRATION—VIBRATION HEARD AND/OR FELT AT CLUTCH PEDAL • CLUTCH—VIBRATION HEARD AND/OR FELT AT CLUTCH PEDAL	Article No. 00-24-1

FORD: 1990-2001 MUSTANG

ISSUE

Some vehicles may exhibit a vibration that may be heard or felt at the clutch pedal between idle and 2300 rpm, and goes away when depressing the clutch at least 25mm (1″). This may be caused by the lack of reserve, quadrant binding, improper clutch cable routing, lack of preload on the release lever, improper or failed release bearing, improper pressure plate-to-flywheel bolt torque, or out-of-parallel finger height on pressure plate.

ACTION

Check, inspect and correct above items as described. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

After verifying clutch pedal vibration condition, inspect/correct the following items:

1. Check for clutch reserve:
   a. Set parking brake.
   b. With shift control selector in third gear, clutch disengaged (pedal down) and engine idling, move shift lever to a position halfway between third gear and Neutral.
   c. Slowly engage clutch (release pedal). The transmission will behave as if in Neutral, but the third gear synchronizer is bypassed.
   d. Gear clash can now be heard if an attempt is made to shift into third gear with the clutch engaged (pedal up), and reserve can now be measured.
   e. While maintaining light pressure on the shift lever, slowly press the clutch pedal to the floor. At a point in the clutch pedal travel, the gear clash will stop and the shift selector will slide easily into third.
   f. At the point when the gear clash stops, press the clutch pedal to the floor.
   g. There must be at least 19mm (0.75″) pedal travel from the point where the gear clash stops and the pedal is pressed to the floor to have adequate reserve. If vehicle has inadequate reserve, replace disc and pressure plate. After replacing the clutch and pressure plate, retest the vehicle to determine if the condition is repaired.
   h. If the condition still exists, proceed to Step 2.
2. To check for proper clutch pedal adjustment, pull up on the clutch pedal to the upstop. Pulling up on the pedal ensures that the quadrant is free to rotate through its complete range of motion against the spring load of the quadrant. This will determine if the quadrant is binding. Cycle the clutch pedal and recheck for pedal vibration.
3. Check for proper clutch cable routing and that the cable isolator grommet is seated in the cowl. Refer to Clutch Controls Section of the appropriate year Mustang Workshop Manual for further details.

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4. To determine if there is proper pre-load on the release lever, put the vehicle on a hoist and remove the dust cover at the transmission. Prop the clutch pedal to the upstop. This releases the pawl from the quadrant. Pull the clutch release lever toward the rear of the vehicle and then release it to determine if there is enough spring and cable tension to seat the release bearing against the pressure plate fingers. While the release lever is in the full forward position against the release bearing, use a spring scale such as 211-034 (T74P-3504-Y) or equivalent (Figure 1), to measure the effort to move the release lever rearward. 1.6 kg (3.5 lb) is the minimum force required to start to move the release lever rearward. If the force is less than 1.6 kg (3.5 lb), and the vehicle passed Steps 2 and 3 above, the quadrant is not functioning properly. Remove the clutch pedal prop. Refer to the appropriate Mustang Workshop Manual Clutch Controls Section for quadrant assembly replacement.
5. If the clutch pedal vibration condition has not been resolved, the transmission must be removed.
6. Check the release lever to assure that the pivot stud fits in the center of the pocket in the lever. If the spring tabs on the release lever are bent, replace the lever.
7. Ensure that release bearing F7ZR-7548-AA is used. The number is stamped on the rear flange of the bearing.
8. Check the pressure plate-to-flywheel attachment bolts for proper torque. Refer to the appropriate Mustang Workshop Manual for proper torque specification for the application.
9. With the pressure plate attached to the flywheel, inspect the finger that rides on the release bearing for a variation in height. If the variation exceeds 1mm (0.040 ″) (low to high), replace the pressure plate. It is necessary to use a dial indicator to measure finger height variation (Figure 2). If replacing the pressure plate, also measure the new plate after installation.

NOTE
WHEN USING DIAL INDICATOR TO CHECK FINGER HEIGHT VARIATION AND FLYWHEEL RUN-OUT, HOLD THE FLYWHEEL IN THE FULL FORWARD OR REARWARD POSITION TO PREVENT MEASUREMENT OF CRANKSHAFT ENDPLAY.

10. Check the flywheel run-out with a dial indicator as follows:
    a. Mount the dial indicator so that the indicator contact point rides on the clutch disc contact surface (Figure 3).
    b. Turn the flywheel; if the run-out exceeds the maximum allowance, 0.2mm (0.008″), install a new flywheel.
11. If replacing the flywheel, re-measure the run-out after installation.
12. Visually check the pilot bearing for damage, unusual wear and improper installation, replace as necessary.
13. Reassemble and test drive the vehicle.

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 505000, 505200, 506000, 597997, 702300, 703100

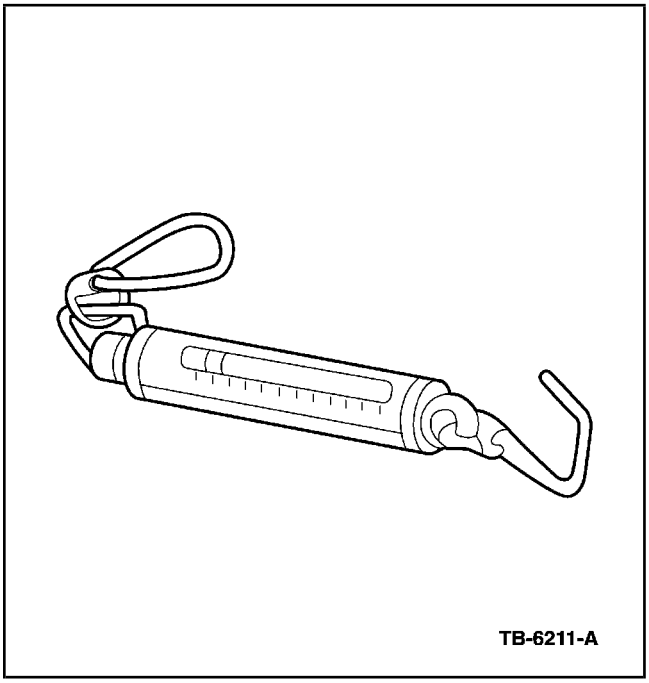

Figure 1 - Article 00-24-1

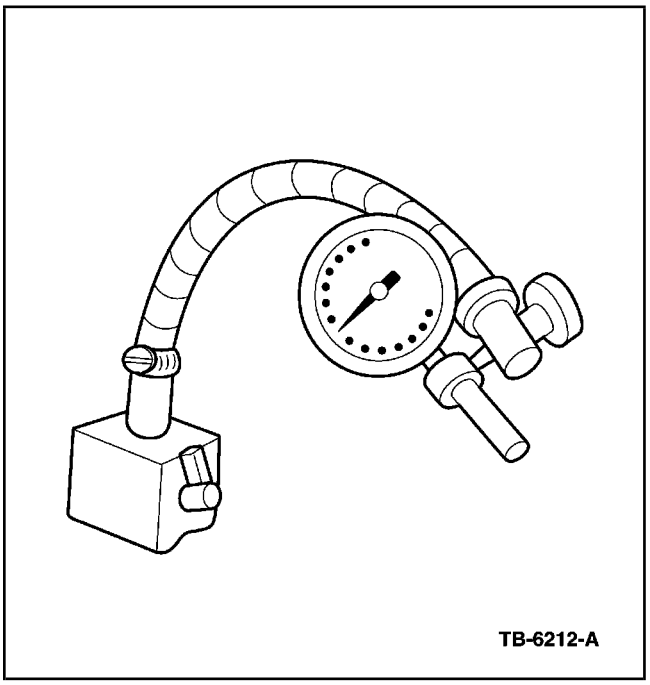

Figure 2 - Article 00-24-1

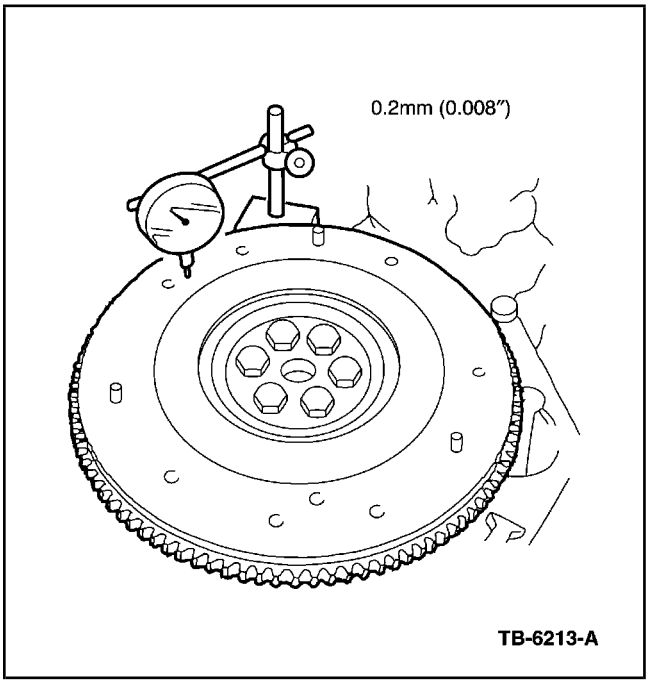

Figure 3 - Article 00-24-1